
TFPO: Token-Level Objective Fusion for Stable Preference Alignment

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Abstract

Preference optimization is usually applied uniformly to entire responses, although explanatory text, final answers, formatting fields, and code need not benefit from the same training signal. We propose TFPO (Token-Fused Preference Optimization), which learns a lightweight gate that routes each response token between a DPO-style preference objective and a chosen-response likelihood anchor. The route is learned without token-level supervision and is stabilized by ratio, smoothness, and entropy regularization. Under a matched ten-benchmark evaluation, TFPO averages 88.00, compared with 83.08 for SimPO and 83.52 for the NLL-anchored SimPO+NLL control (three-seed means). It also improves external alignment across Qwen, Llama, and Mistral backbones and is effective in multimodal preference tuning. Under repeated sampling, TFPO raises answer agreement and majority-answer accuracy while preserving explanation diversity. On 1,000 gate- and model-blind annotated responses from five extractable-answer tasks, its content gate recovers answer-bearing spans with 0.85 AUPRC, versus 0.55 for the strongest strict position-only control. These results support token-level objective routing as a practical way to improve both preference alignment and answer stability.

1 Introduction

Preference optimization has become a standard post-training stage for large language models (LLMs) and multimodal large language models (MLLMs). Methods such as DPO [30], SimPO [28], and ORPO [18] offer efficient alternatives to reinforcement learning from human feedback. Yet most preference objectives still treat a completion as a homogeneous sequence that receives one scalar preference decision. This creates a coarse credit-assignment problem for a functionally heterogeneous output.

LLM responses are not homogeneous token strings. Their spans range from explanatory prose and answer-bearing fields to formatting delimiters, schema keys, and executable code. These regions should not necessarily receive the same optimization signal. Reasoning tokens can benefit from preference shaping: they determine whether an explanation is helpful, concise, and well justified. In contrast, final answer fields, option letters, short numeric answers, JSON keys, or code fragments often need likelihood anchoring to remain parseable and stable. When all tokens inherit the same sequence-level preference pressure, these roles can interfere: the model may learn a more preferred style while destabilizing the small span that actually determines correctness.

We introduce **Token-Fused Preference Optimization (TFPO)**, a token-level objective routing framework for this functional heterogeneity. TFPO learns a small gate over response tokens and uses it to route tokens between two objectives: a preference objective for reasoning-sensitive regions and a chosen-response likelihood anchor for answer-bearing regions. The method does not require explicit labels for reasoning or conclusion spans. Instead, the gate learns a latent soft partition that minimizes the fused objective.

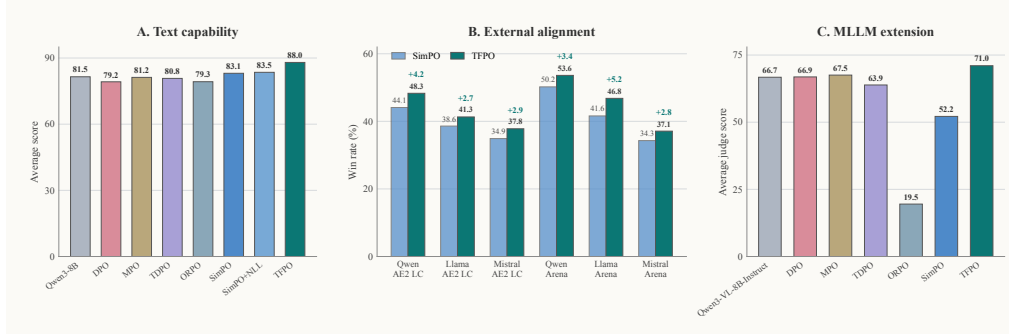


Figure 1: Summary of the main empirical picture. Left: TFPO improves the main ten-benchmark average over both SimPO and the NLL-anchored SimPO+NLL control on Qwen3-8B. Middle: TFPO consistently improves over SimPO on judge-based alignment benchmarks across Qwen, Llama, and Mistral backbones. Right: under multimodal preference tuning, TFPO yields the best average among the compared methods.

The core claim is not merely that mixing preference and likelihood is helpful. Existing mixed-loss recipes already suggest that it can be. Our claim is sharper: *where* each objective is applied matters. TFPO treats preference alignment as token-level functional credit assignment rather than uniform sequence-level credit assignment. A token-wise routing mechanism can reduce interference between preference shaping and answer anchoring, preserving final-answer stability while still improving alignment quality. Figure 2 gives the visual overview of this routing view.

Our experiments evaluate both the empirical reach of TFPO and the behavior of its learned route. Across text capability, cross-family alignment, and multimodal preference tuning, TFPO improves strong preference-optimization baselines; targeted routing analyses and repeated sampling support content-sensitive allocation and more stable answers.

The broader diagnostic lesson is that smooth losses and improving sequence-level rewards are not sufficient evidence of healthy preference alignment. They can hide routing failures when answer-bearing tokens lose their likelihood anchor. Preference optimization should ask not just whether reward improves, but where optimization pressure is applied within the response.

Contributions. We make three contributions:

1. We formulate stable preference alignment as a token-level functional credit-assignment problem and instantiate it as TFPO, a routing method between preference optimization and likelihood anchoring.
2. We introduce a lightweight gate regularization recipe—ratio, smoothness, and entropy control—that controls route mass, local switching, and premature saturation while remaining easy to implement.
3. We provide broad empirical evidence across text, cross-family external alignment, and multimodal settings, together with strict position controls, gate-blind gold-span annotations, counterfactuals, and repeated-sampling stability tests that directly probe the routing claim.

2 Related Work

Preference alignment with learned rewards. Early RLHF pipelines train a reward model from human comparisons and optimize the policy with reinforcement learning, most commonly PPO-style updates [31, 8, 33, 29]. RLAIIF and constitutional variants reduce direct human labeling with AI feedback or rule-based principles [5]. Recent online RL work revisits the update rule itself, from REINFORCE-style simplifications to GRPO/RLVR-style reasoning pipelines [1, 32, 15, 39]. TFPO is complementary: it uses offline preference pairs, but shares the view that scalar rewards can hide unstable credit assignment.

Sequence-level preference optimization. DPO [30] converts preference learning into a logistic classification problem over chosen and rejected responses relative to a frozen reference model. Rank-

Table 1: Positioning of TFPO among preference-optimization methods. The key distinction is not whether a method uses preference data, but whether it can assign different objectives to different token roles inside the same response.

Method	Granularity	Token-level mechanism	Reference model?	Likelihood / NLL anchor?
DPO	Sequence-level	None	Yes	No
SimPO	Sequence-level	None	No	No
ORPO	Sequence-level	None	No	Yes, chosen NLL
MPO-style	Sequence-level	Mixed objectives on full sequence	Yes	Yes, full-response SFT/NLL
TDPO	Token-level	Tokenized preference objective	Yes	No
TIS-DPO	Token-level	Token importance weighting inside DPO	Yes	No
TFPO	Token-level	Functional objective routing via gate	Yes	Yes, token-routed chosen NLL

based and direct alternatives, including SLiC-HF [45], RRHF [41], IPO [3], KTO [13], WPO [48], SimPO [28], and ORPO [18], change the surrogate, reference usage, data weighting, or normalization strategy. These methods are important because they show that the alignment objective can be simplified substantially. However, they still assign credit at response level: all tokens in a completion inherit the same sequence-level preference decision.

Mixed-objective preference training. Mixed-loss recipes combine preference optimization with auxiliary generation losses. MPO-style training in recent multimodal reasoning work [37], for example, mixes preference and SFT-style objectives to reduce the instability of preference-only tuning. TFPO shares the motivation that preference optimization can disturb answer quality, but differs by applying objective mixing at token granularity rather than uniformly across the whole completion.

Token-level preference modeling. TDPO [43] and TIS-DPO [24] both move beyond treating the response as an atomic unit. TDPO formulates token-level preference optimization, while TIS-DPO estimates token importance weights inside a preference loss. TFPO is complementary but structurally different: it does not only reweight tokens within a preference objective; it routes tokens *between* preference and likelihood objectives.

Stability under alignment. Preference tuning can improve style or helpfulness while hurting exact answer formatting, output schemas, or short-answer correctness. We use *stability* to mean reduced answer/format drift under preference training, especially for final answer fields that are easy to parse and verify. This framing is central to TFPO: the method is designed to preserve stable conclusions while leaving reasoning tokens available for preference shaping. One alternative is to identify final-answer spans with task-specific parsers or templates, but such supervision is brittle across open-ended, code, schema, and multimodal settings. TFPO instead learns a latent route from preference data without requiring token-level span labels.

Table 1 summarizes the gap that TFPO targets. Prior work either improves the sequence-level preference surrogate, mixes objectives at full-response granularity, or reweights tokens inside a preference loss. TFPO instead treats the output as functionally heterogeneous: different token roles can be assigned to different objectives inside the same response.

3 Method

3.1 Setup

Let x be a prompt, y^+ a preferred response, and y^- a rejected response. We train a policy model π_θ initialized from an instruction-tuned checkpoint, with a frozen reference model π_{ref} when using the reference-based objective. Standard DPO forms a preference margin from the full log-probability of each response. This is elegant, but it also binds heterogeneous token roles to one scalar preference decision: a final numeric answer, a chain-of-thought explanation, a markdown delimiter, and a code token all contribute to the same sequence-level margin.

TFPO changes this credit-assignment step. It keeps the familiar preference-pair setup, but treats the response as a mixture of functional token roles and asks the model to learn a soft route over them. Tokens that look like reasoning, elaboration, or preference-sensitive style are routed toward the preference objective. Tokens that look like final answer fields, schema fields, fragile output formats, or code/output tokens are routed toward a chosen-response likelihood anchor. The partition

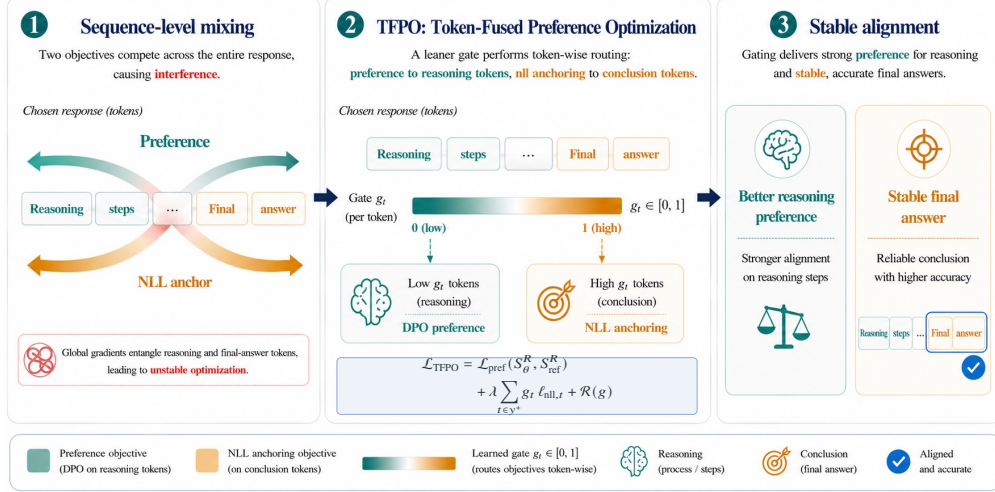


Figure 2: Overview of TFPO. Instead of assigning one optimization signal to a functionally heterogeneous response, TFPO learns token-wise routing: preference gradients primarily shape reasoning-sensitive tokens, while likelihood anchoring stabilizes answer-bearing and conclusion-sensitive tokens. The displayed loss is schematic; the actual implementation uses the routed DPO sequence scores in Eqs. (2)–(4) and the full objective in Eq. (9).

is latent: no training example tells the model which tokens are “reasoning” or “conclusion” tokens. In this sense, TFPO is not simply a token weighting variant; it is an instance of token-level objective routing for functionally heterogeneous completions.

3.2 Token gate

For each response token, TFPO predicts a scalar gate

$$g_t = \sigma(\text{sg}(\mathbf{h}_t)\mathbf{W}_g + b_g) \in (0, 1), \quad (1)$$

where $\mathbf{h}_t$ is the decoder hidden state, $\text{sg}(\cdot)$ denotes stop-gradient, and $\mathbf{W}_g, b_g$ are a lightweight gate head. We use the convention that a large g_t routes token t toward the conclusion / likelihood objective, while a small g_t routes it toward the reasoning / preference objective.

Two design choices are worth spelling out. First, the default gate is deliberately small: a single linear head over hidden states. This makes it cheap and limits the risk that the gate becomes a second model. Second, we stop gradients through $\mathbf{h}_t$ on the gate path. The gate can read the model state, but the backbone cannot reshape hidden states merely to escape one part of the objective; Appendix A gives the exact computational graph.

3.3 Token-level objective fusion

From sequence scores to token-routed scores. The SFT/NLL loss is naturally token-level: each token has a local negative log-likelihood. DPO, by contrast, first sums token log-probabilities into one sequence score and then applies a pairwise sigmoid. TFPO keeps this DPO structure, but replaces the unweighted sequence sum with a token-weighted sum. Define conclusion and reasoning weights as $w_t^C = g_t$ and $w_t^R = 1 - g_t$. For any response y , the reasoning-weighted log-probability is

$$S_\theta^R(y | x) = \sum_{t=1}^{T_y} (1 - g_t) \log \pi_\theta(y_t | x, y_{<t}). \quad (2)$$

Tokens with low gate values contribute strongly to this score; tokens with high gate values are mostly excluded from the preference comparison. This is the central move of TFPO: it does not change the preference-pair data format, but it changes which functional regions are responsible for the preference margin.

Preference objective. For the preference term, we replace the four full-response log-probabilities in DPO with the reasoning-weighted scores:

$$\Delta_{\theta}^{TFPO} = [S_{\theta}^R(y^+ | x) - S_{\theta}^R(y^- | x)] - [S_{\text{ref}}^R(y^+ | x) - S_{\text{ref}}^R(y^- | x)], \quad (3)$$

and the preference loss is

$$\mathcal{L}_{\text{pref}}^{TFPO} = -\log \sigma(\beta \Delta_{\theta}^{TFPO}). \quad (4)$$

This preference term is therefore not a sum of independent token-level preference losses. The gate first forms reasoning-weighted sequence scores, and those scores then enter the same DPO-style pairwise sigmoid. The chosen and rejected responses each receive their own gate values from their own hidden states; there is no token-to-token alignment assumption between them.

We reuse the policy-side weights for the corresponding reference log-probabilities and treat those weights as constants for the frozen reference model. In implementation, the reused policy-side gates are detached when multiplying reference log-probabilities, so the reference score does not introduce additional gradients through the frozen reference term. This keeps policy and reference scores in the same routed token subspace. If the policy score were routed but the reference score remained unweighted, conclusion tokens could leak back into the preference margin through the reference term.

Likelihood objective. The likelihood term provides the complementary signal. It is computed only on the chosen response:

$$\mathcal{L}_{\text{nll}}^{TFPO} = \sum_{t=1}^{T_{y^+}} g_t \left(-\log \pi_{\theta}(y_t^+ | x, y_{<t}^+) \right). \quad (5)$$

This chosen-side NLL term anchors tokens that should remain stable, such as final answers, schema fields, and code-like output fragments. For chosen responses, the gate allocates token mass between the routed preference margin and the likelihood anchor. Rejected tokens participate only in the routed preference margin and are never imitated.

3.4 Regularization and full objective

The objective above admits trivial shortcuts. If every token receives the preference objective, TFPO reduces toward token-weighted DPO and loses answer anchoring. If every token receives the NLL objective, it becomes mostly imitation and loses preference learning. If the gate changes every token, the routing becomes hard to interpret and unstable. We therefore use three simple regularizers.

Ratio regularization. Let $\bar{g} = \frac{1}{T} \sum_t g_t$. The ratio constraint keeps the conclusion-side mass inside a broad range:

$$\mathcal{L}_{\text{len}} = [\max(0, r_{\min} - \bar{g})]^2 + [\max(0, \bar{g} - r_{\max})]^2. \quad (6)$$

The bounds are intentionally loose: they control route mass without prescribing where conclusion tokens should be routed.

Smoothness regularization. The smoothness penalty is a total-variation term:

$$\mathcal{L}_{\text{tv}} = \sum_{t=2}^T |g_t - g_{t-1}|. \quad (7)$$

This encourages contiguous spans rather than salt-and-pepper token assignments, which also makes the learned routing easier to inspect in heatmaps.

Entropy regularization. Finally, we maximize a small Bernoulli entropy term:

$$\mathcal{L}_{\text{ent}} = \frac{1}{T} \sum_{t=1}^T [-g_t \log g_t - (1 - g_t) \log(1 - g_t)]. \quad (8)$$

The entropy term discourages premature hard decisions while the two objectives are calibrating.

Full objective. The full objective is

$$\mathcal{L}^{TFPO} = \lambda_{\text{pref}} \mathcal{L}_{\text{pref}}^{TFPO} + \lambda_{\text{nll}} \mathcal{L}_{\text{nll}}^{TFPO} + \lambda_{\text{len}} \mathcal{L}_{\text{len}} + \lambda_{\text{tv}} \mathcal{L}_{\text{tv}} + \lambda_{\text{ent}} \mathcal{L}_{\text{ent}}. \quad (9)$$

Our default recipe uses $\beta = 0.1$, $\lambda_{\text{pref}} = 1$, $\lambda_{\text{nll}} = 0.2$, $[r_{\min}, r_{\max}] = [0.05, 0.9]$, with TV and entropy regularization enabled. The most useful way to read this objective is as a soft mixture-of-objectives over tokens: the model is not told where the final answer begins, but it is pressured to discover a stable allocation because the two objectives reward different token roles. Appendix A gives the exact computational graph and implementation notes.

4 Experiments

4.1 Experimental setup

Models and training data. The main comparison uses the post-trained Qwen/Qwen3-8B checkpoint [38]. All preference methods are trained on the same cleaned UltraFeedback-style preference split [10] with matched training settings; exact run details are in Appendix B and Table 6. Appendix C reports robustness on a larger 2M preference mixture; because its composition also changes, we do not treat it as controlled data scaling. Cross-family checks use Llama-3.1-8B-Instruct [11] and Mistral-7B-Instruct-v0.3 [19] under the same external alignment protocol. The multimodal extension trains preference methods from Qwen3-VL-8B-Instruct [4].

Baselines. We compare TFPO with DPO, MPO-style mixed preference optimization, SimPO, TDPO, and ORPO. The off-the-shelf model is included when it is the natural initialization baseline.

Capability benchmarks. We evaluate on ten benchmarks spanning reasoning, instruction following, QA/exam tasks, math, and code: BBH [35], GSM8K [9], HumanEval [7], IFEval [47], MMLU [16], RACE [21], C3 [34], Gaokao [44], MATH [17], and MBPP [2]. The MATH column uses EleutherAI/hendrycks_math, not MATH-500; Appendix B reports a separate standard MATH-500 bridge. The main aggregate is the arithmetic mean across the ten benchmark scores. We use one frozen, method-independent prompt, decoding, extraction, and scoring protocol per benchmark; all fine-tuned values in the capability table are equal-weight means over three training seeds. Generative tasks use Qwen3’s non-thinking sampling configuration and a 32,768-token output cap; Appendix B records the per-task sample counts, completeness audit, and the separate thinking/non-thinking MATH-500 bridge.

External alignment benchmarks. For AlpacaEval 2.0 [12], we generate model outputs in the official format and evaluate with the official length-controlled annotator configuration `weighted_alpaca_eval_gpt4_turbo_new`. For Arena-Hard-v2.0-Preview [22], we generate official-format answers and use the official GPT-4.1 judge with markdown and length controls.

Multimodal benchmarks. For Qwen3-VL-8B-Instruct, we evaluate ten multimodal benchmarks: MathVista [27], MathVision [36], MMMU [42], AI2D [20], MMVet [40], LLaVA-Bench [25], MM-Bench [26], MMStar [6], POPE [23], and HallusionBench [14]. We use `gpt-4o-mini` judge-based scoring and report the VLM Score for LLaVA-Bench rather than the relative score.

4.2 Main capability results

Figure 1 gives the high-level result summary before the detailed tables. Table 2 is the primary capability comparison. Under the matched protocol, TFPO is strongest on the ten-benchmark average and on every constituent benchmark. The broad SimPO comparison and the NLL-anchored SimPO+NLL comparison separate the overall performance claim from the question of whether likelihood anchoring alone explains the gain.

The gain is not a single-benchmark artifact: TFPO improves instruction following, math, general academic reasoning, reading/exam-style tasks, and code generation. It exceeds the strongest broad baseline, SimPO, by 4.92 points and the NLL-anchored SimPO+NLL control by 4.48 points. Thus likelihood anchoring is useful but does not recover the benefit of assigning preference and anchoring signals to different token locations.

Table 2: Matched capability evaluation on post-trained Qwen3-8B. Fine-tuned rows are three-seed means under one frozen, method-independent benchmark protocol. Best values are bolded and second-best values are underlined.

Method	Avg.	BBH	GSM8K	HumanEval	IFEval	MMLU	RACE	C3	Gaokao	MATH	MBPP
Qwen3-8B	81.50	79.2	92.1	78.4	83.2	79.5	84.1	88.5	70.2	83.7	76.1
DPO	79.22	78.4	88.2	78.0	80.3	75.2	84.5	87.1	66.2	81.5	72.8
MPO	81.22	79.4	89.2	84.1	80.9	75.0	86.0	88.0	70.4	83.0	76.2
SimPO	83.08	80.5	92.4	86.1	82.1	78.1	88.0	91.2	70.6	84.3	77.5
TDPO	80.76	79.8	90.6	82.0	81.3	76.0	85.3	89.0	68.3	82.1	73.2
ORPO	79.26	78.4	88.0	80.3	79.6	75.0	84.0	88.6	66.0	80.7	72.0
SimPO+NLL	83.52	80.5	92.5	86.5	82.3	79.0	88.5	91.0	71.4	85.3	78.2
TFPO	88.00	86.0	96.1	91.3	88.1	84.6	91.0	94.3	76.0	89.6	83.0

Table 3: Cross-family external alignment results. AlpacaEval 2.0 uses the official length-controlled weighted GPT-4-Turbo annotator configuration; Arena-Hard-v2.0-Preview uses the official GPT-4.1 judge with markdown and length controls. For AlpacaEval, LC and WR denote length-controlled and raw win rate; Arena-Hard WR is the official control-adjusted win rate.

Method	🦾 Qwen3-8B			🦙 Llama-3.1-8B-Instruct			🐼 Mistral-7B-Instruct-v0.3		
	AlpacaEval 2.0		Arena-Hard	AlpacaEval 2.0		Arena-Hard	AlpacaEval 2.0		Arena-Hard
	LC	WR	WR	LC	WR	WR	LC	WR	WR
SFT/Instruct	19.8	31.2	34.1	7.1	23.1	27.5	11.3	19.2	13.8
DPO	24.1	34.6	38.0	10.6	25.7	30.1	12.6	22.7	19.8
MPO	30.8	37.2	41.5	23.4	28.8	32.4	17.4	25.0	20.1
SimPO	44.1	49.7	50.2	38.6	44.9	41.6	34.9	39.1	34.3
TDPO	36.9	41.6	42.1	27.1	38.2	35.3	25.1	36.2	25.7
ORPO	17.4	19.7	36.1	9.4	8.7	28.6	10.1	12.5	14.9
TFPO	48.3	56.1	53.6	41.3	48.6	46.8	37.8	41.0	37.1

4.3 External alignment and cross-family transfer

Table 3 tests whether the gains persist on open-ended judge-based alignment benchmarks and across model families. TFPO is strongest on every reported AlpacaEval 2.0 and Arena-Hard-v2.0-Preview column. This matters because AlpacaEval 2.0 length control reduces verbosity bias, while Arena-Hard-v2.0-Preview is a harder pairwise benchmark with official length and markdown controls.

The cross-family pattern is important for the method claim. The same routing recipe improves Qwen, Llama, and Mistral backbones, suggesting that the gain is not tied to one tokenizer, one instruction-tuning recipe, or one model family. Appendix E reports the three-seed robustness check, while Appendix C separately reports external alignment on the larger preference mixture.

4.4 Core routing evidence

The central question is whether TFPO benefits merely from adding chosen-side NLL or whether its content-dependent route is functionally different from a position shortcut. Table 2 answers the first question under the matched capability protocol. Table 4 answers the second with three direct routing evaluations that are independent of the capability-scoring pipeline. RR-Preserve measures correct-answer retention after removing low-score reasoning tokens; CR-Failure measures answer failure after removing high-score answer tokens.

The strict position-only controls read only normalized position and response-length features. Across equal deletion budgets, the content gate improves over the stronger position MLP by 12.3 RR-Preserve and 21.5 CR-Failure points. Under method-independent gold removal on each variant’s own responses, its joint score is 84.4 versus 69.9. On the separate gold-recovery evaluation, it improves by 0.30 AUPRC (95% response-bootstrap CI [0.27, 0.33]) and 0.22 F1 ([0.19, 0.25]), with positive AUPRC differences on all five tasks. Appendix D additionally reports a *content+position* auxiliary-feature variant; because it reads hidden-state content, it is not used as position-only evidence.

Table 4: Core routing evidence against strict position-only controls. Panel A averages equal-token-budget interventions over 10–40% removal budgets. Panel B uses method-independent gold spans for removal. Panel C evaluates gate scores against blind answer-span annotations on five extractable-answer tasks.

A. Equal-removal-budget intervention (mean over 10–40% budgets)				
Router	RR-Preserve $\uparrow$	CR-Failure $\uparrow$	Δ RR vs. MLP	Δ CR vs. MLP
Pure-position linear	68.9	41.2	–	–
Pure-position MLP	72.6	46.0	0.0	0.0
TFPO content gate	84.9	67.5	+12.3	+21.5
B. Method-independent gold-span intervention				
TFPO variant	Reasoning preserve $\uparrow$	Answer failure $\uparrow$	Joint geom. $\uparrow$	Δ Joint vs. MLP
Pure-position linear route	73.1	59.1	65.7	–
Pure-position MLP route	76.5	63.8	69.9	0.0
Content-gated route	88.6	80.4	84.4	+14.5
C. Gold answer-span recovery (1,000 responses)				
Router	AUPRC $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F1 $\uparrow$
Pure-position linear	0.50	0.58	0.52	0.55
Pure-position MLP	0.55	0.63	0.57	0.60
TFPO content gate	0.85	0.84	0.80	0.82

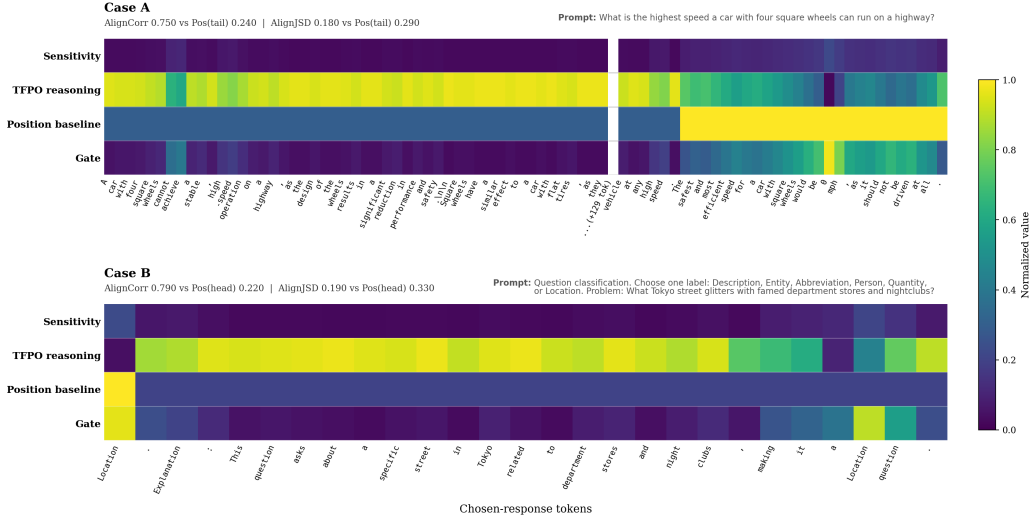


Figure 3: Held-out token-routing examples. *Sensitivity* denotes the normalized absolute policy-reference log-probability difference $|\Delta \log p|$. The other rows show TFPO reasoning allocation, a matched-budget position heuristic, and the learned gate; Case B places the answer label first.

4.5 Mechanistic evidence

The benchmark results show that TFPO works; the next question is whether its route is content-sensitive and whether that route produces the intended stability. We test these predictions with gold answer spans, answer-position counterfactuals, repeated sampling, and method-independent token-removal interventions.

Gold spans and position controls. Two annotators, blind to gate values and model identity, label attempted answer-bearing spans independently on 300 responses before adjudication ($\kappa = 0.92$, span IoU = 0.89). The final evaluation contains 1,000 responses (200 from each of five extractable-answer tasks) after 18 prespecified, score-blind integrity exclusions. The 1,000-response gold labels are used only for evaluation; AUPRC is the primary threshold-free metric, and the F1 threshold is fixed once on disjoint development data. Table 4 shows that TFPO reaches 0.85 AUPRC versus 0.55 for the strongest strict position-only MLP. In a content-preserving counterfactual that moves the same answer to the beginning, middle, or end, TFPO remains at 0.83/0.84/0.85 AUPRC, whereas a pure-position linear router varies from 0.25 to 0.62. Thus the gate follows answer content rather than a fixed tail shortcut.

Table 5: Repeated-sampling stability, macro-averaged over BBH, GSM8K, and MATH and then over three training seeds. All values use the same frozen answer canonicalizer and include missing/invalid generations in the denominator.

Method	Agreement@4↑	MajorityAcc@4↑	AvgAcc@4↑	Div.@4↑	Valid↑
SimPO	88.33	80.69	85.73	59.01	99.67
SimPO+NLL	89.01	81.69	86.10	59.72	99.70
TFPO	92.69	86.32	90.57	62.33	99.80

Figure 3 gives qualitative route examples. Its sensitivity row is descriptive; semantic evaluation uses the independently annotated gold spans in Table 4.

Final-answer stability and explanation diversity. For BBH, GSM8K, and MATH, we draw $K = 4$ matched non-thinking responses per prompt for each of three training seeds. AnswerAgreement@4 is the modal canonical-answer fraction, MajorityAcc@4 evaluates the unique plurality answer against gold, AvgAcc@4 averages per-generation correctness, and ExplanationDiv@4 measures pairwise diversity after removing final-answer fields. Appendix A gives the edge-case rules. Tables abbreviate AnswerAgreement, ExplanationDiv, and valid-answer rate as Agreement, Div., and Valid.

Table 5 shows the intended signature: relative to SimPO+NLL, TFPO improves answer agreement by 3.68 points (paired-prompt 95% CI [2.38, 4.98]), majority accuracy by 4.63 points, and average accuracy by 4.47 points, while increasing rather than sacrificing explanation diversity. The agreement gain is positive for all three training seeds. Appendix E gives per-task and seed-level results, a MATH-500 robustness check, strict-position threshold/equal-budget sweeps, and method-independent gold interventions.

4.6 Multimodal extension

Multimodal preference tuning amplifies the need for token-level routing. Many VLM benchmarks combine visual reasoning with fragile answer fields: OCR short answers, counting numbers, chart/table options, multiple-choice letters, structured VQA fields, or yes/no hallucination judgments. TFPO is therefore a natural fit: visual reasoning tokens can be preference-shaped, while final answers and schema-like tokens remain likelihood anchored. As summarized in Figure 1, TFPO reaches the strongest multimodal average (71.03), improving over MPO by 3.49 points and over Qwen3-VL-8B-Instruct by 4.29 points. Appendix F gives the full table and qualitative examples of format drift, wrong option selection, and excessive verbosity.

5 Limitations and Conclusion

Limitations and broader impact. The main capability comparison (including the NLL-anchored control), repeated-sampling analysis, and TFPO–SimPO cross-family comparisons use three training seeds. Gold-span semantics are evaluated only on five tasks with extractable answer fields and should not be read as a unique partition for arbitrary open-ended text. The learned route is latent and not necessarily identifiable, and our model-scale evidence is limited to 7–8B LoRA settings. Judge-based evaluations remain judge-dependent. TFPO can improve parseability and answer stability, but it does not remove risks such as hallucination, bias, or misuse.

Conclusion. TFPO reframes preference optimization as token-level credit assignment: choosing which response parts should receive preference pressure and which should retain likelihood anchoring. The capability gains over an NLL-anchored control, together with routing improvements over strict position-only routers, blind gold-span recovery, answer-position counterfactuals, and repeated-sampling stability, support the importance of *where* objectives act inside a response. Cross-family and multimodal results further show the empirical reach of this routing view.

Acknowledgments and Disclosure of Funding

This work received no external funding. The author declares no competing interests.

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Appendix

- Appendix A: Method Details and Derivations
- Appendix B: Experimental Details and Reproducibility
- Appendix C: Robustness to a Larger Preference Mixture
- Appendix D: Ablations and Training Diagnostics
- Appendix E: Full Mechanistic Evaluation Results
- Appendix F: MLLM Evaluation Details and Qualitative Examples

A Method Details and Derivations

Exact routed computation. The policy and reference scores use the same numerical routing weights but different gradient paths. For either chosen or rejected response y , the implementation computes

$$S_{\theta}^R(y \mid x) = \sum_t (1 - g_t) \log \pi_{\theta}(y_t \mid x, y_{<t}),$$

$$S_{\text{ref}}^R(y \mid x) = \sum_t \text{sg}(1 - g_t) \log \pi_{\text{ref}}(y_t \mid x, y_{<t}),$$

where sg denotes stop-gradient. These four sequence scores enter the single DPO-style sigmoid in Eqs. (3)–(4). The reference model is frozen and the detached reference weights do not update the policy gate. The chosen-only NLL branch is the weighted token sum in Eq. (5); rejected tokens are never imitated. Regularizers are computed separately on chosen and rejected gates and then averaged.

Descriptive policy–reference diagnostic. For the descriptive diagnostic in Figure 3, define $\delta_t = \log \pi_{\theta}(y_t \mid x, y_{<t}) - \log \pi_{\text{ref}}(y_t \mid x, y_{<t})$. AlignCorr and AlignJSD compare the normalized $|\delta_t|$ map with the normalized reasoning allocation $(1 - g_t)$. We use AlignCorr only to describe policy–reference shift; semantic correspondence is evaluated separately against gold spans.

Gold answer-span evaluation. Gold labels mark the span in which a response attempts to express the task-schema answer, independent of whether that attempted answer is correct. Two annotators are blind to model identity and all gate scores. AUPRC is the primary threshold-free measure; the F1 threshold is selected once on disjoint development data. The evaluation is restricted to GSM8K, MATH, RACE, ARC-Challenge, and BoolQ, where an attempted answer-bearing span is operationally extractable. It does not assume a unique semantic partition for arbitrary open-ended responses.

Stability and interventions. For canonical answer equivalence $f(\cdot)$ and K generations, Answer-Agreement is

$$\text{Agreement}@K(x) = \max_{a \in \mathcal{A}_{\text{valid}}} \frac{1}{K} \sum_{k=1}^K \mathbb{I}[f(y^{(k)}) = a].$$

Here $\mathcal{A}_{\text{valid}}$ contains only valid canonical answers. Missing/invalid generations remain in the denominator but cast no vote; an all-invalid prompt has agreement zero. MajorityAcc is one only when a unique plurality answer is gold-equivalent (ties are zero); AvgAcc averages the K correctness indicators. Explanation diversity is the mean pairwise Jaccard distance after removing final-answer fields. For interventions, RR-Preserve asks whether deleting low-score reasoning tokens preserves a correct answer, whereas CR-Failure asks whether deleting high-score answer tokens breaks it. We report threshold and equal-token-budget sweeps using each router’s own scores, plus method-independent gold-span removal; no sequence-level baseline is evaluated with a shared TFPO segmenter.

Implementation notes. Before routing, the implementation applies the same deterministic short-response safeguard to every variant. For a response of length $T \geq 2$, it clamps the response mean $\bar{g}$ to $[1/T, 1 - 1/T]$, shifts every response-token gate by the resulting mean difference, and clips only for numerical safety. Thus each objective retains approximately one token of soft mass even when the ratio penalty is weak on a short response. The gate head is attached as lightweight trainable state before LoRA wrapping. TFPO disables padding-free RLHF and precomputed reference log-probabilities because the reference terms must be dynamically reweighted by the policy gate. SFT replay, when enabled, bypasses the gate and contributes only plain NLL.

B Experimental Details and Reproducibility

Training split. The main experiments use 60k cleaned UltraFeedback-style preference pairs. We keep this as the main narrative setting because it provides a controlled comparison across methods while keeping training inexpensive enough for ablations and repeated mechanism evaluations. The larger 2M preference mixture is reported separately as a robustness check; it changes both sample count and composition and is therefore not a controlled scaling experiment.

Compute. All reported runs use NVIDIA A100-80GB GPUs. The main setup runs for one epoch on 8 GPUs and takes roughly 3 hours (24 A100 GPU-hours per run); the larger 2M mixture runs for one epoch on 16 GPUs and takes roughly 50 hours (800 GPU-hours per run). In the matched 60k efficiency audit, TFPO uses 3.00 hours and 66.16 GiB peak memory per GPU versus DPO’s 2.83 hours and 63.0 GiB—a $1.06\times$ time and $1.05\times$ memory overhead. Reference-free SimPO remains cheaper, as expected. Mechanism evaluation reuses the same checkpoint family and runs offline on cached benchmark subsets.

Table 6: Frozen run card for the main TFPO LoRA setting and its evaluations. Baselines use the shared launcher settings unless the method-specific objective requires otherwise.

Item	Setting
Optimizer / schedule	AdamW through the MS-SWIFT trainer [46]; learning rate 1×10^{-4} ; cosine scheduler; warmup ratio 0.05.
Batching	Eight GPUs; LoRA micro-batch 2 per GPU with gradient accumulation 2, giving effective global batch size 32.
Length / precision	Maximum sequence length 4096; bfloat16 training; gradient checkpointing enabled.
LoRA	Rank 8, alpha 32, dropout 0.05, target modules <code>all-linear</code> .
Training preprocessing	<code>qwen3_nothinking</code> template with <code>add_non_thinking_prefix=false</code> ; the same tokenizer and preprocessing are used across methods.
Training horizon	One epoch; evaluation and checkpointing every 10% of the epoch; step 0 is evaluated for diagnostics but is not a selection candidate; save limit 100.
Search budget	Broad methods receive the same nine-configuration search budget; NLL-anchored controls use one fixed configuration over three seeds. TDPO retains its prescribed extra SFT stage, so search counts rather than total training stages are matched.
Validation / checkpoint selection	A fixed set of 2,000 preference pairs, disjoint from training and reported tests, selects among the ten checkpoints at 10%, 20%, ..., 100%. Within each method and seed, the selector is $0.5 \text{ rank}(\text{pairwise accuracy}) + 0.5 \text{ rank}(-\text{chosen-token-mean NLL})$, using percentile ranks over those ten candidates; exact ties select the earlier checkpoint. Reported test benchmarks never enter selection.
TFPO objective	$\beta = 0.1$, $\lambda_{\text{pref}} = 1.0$, $\lambda_{\text{nll}} = 0.2$, $\lambda_{\text{len}} = 1.0$, $\lambda_{\text{tv}} = 0.02$, entropy init $\lambda_{\text{ent}} = 0.01$.
Gate regularization	Linear gate with stop-gradient; $r_{\text{min}} = 0.05$, $r_{\text{max}} = 0.9$; entropy weight decays linearly from 0.01 to 0 over the first 35% of training. The short-response safeguard keeps approximately one token of soft mass in each branch.
NLL-anchored controls	DPO+NLL uses $\beta = 0.1$ and <code>rpo_alpha=0.2</code> ; SimPO+NLL uses $\beta = 2.0$, $\gamma = 1.0$, and <code>cpo_alpha=0.2</code> . Both add chosen-response token-mean NLL with coefficient 0.2 and contain no gate, ratio, TV, or entropy term.
Reference handling	Reference-based methods use a frozen reference model; SimPO, SimPO+NLL, and ORPO are reference-free. In TFPO, policy-side reasoning weights are reused for reference log-probabilities and detached in the reference term.
Capability decoding	Qwen3 non-thinking mode; temperature 0.7, $\text{top-}p = 0.8$, $\text{top-}k = 20$, $\text{min-}p = 0$, presence penalty 1.5, max new tokens 32,768. Table 2 reports AvgAcc@4 for BBH/GSM8K/MATH ($n = 4$) and the standard unbiased pass@1 estimator for HumanEval/MBPP ($n = 20$); remaining benchmark protocols use $n = 1$.
External alignment decoding	AlpacaEval 2.0: temperature 0, $\text{top-}p = 1$, max new tokens 1024; Arena-Hard-v2.0-Preview: temperature 0, $\text{top-}p = 1$, max new tokens 4096.
Mechanism decoding	Stability: $K = 4$ matched non-thinking samples on BBH/GSM8K/MATH for each of three training seeds. Interventions report a 0.3–0.7 threshold sweep and equal-token budgets of 10–40%, rather than relying on one threshold.
MATH-500 bridge	Standard 500-example test split, $n = 8$ aligned samples, mode-specific Qwen3 decoding, and max new tokens 32,768; shorter caps are evaluated only in the length-sensitivity audit.

External benchmark workflow. We generate AlpacaEval 2.0 and Arena-Hard-v2.0-Preview model outputs in their official answer formats and evaluate them with the corresponding public judging pipelines. AlpacaEval uses `weighted_alpaca_eval_gpt4_turbo_new`; Arena-Hard-v2.0-Preview uses the official GPT-4.1 judge with markdown and length controls. We report AlpacaEval length-controlled win rate (LC) and raw win rate (WR), and the official control-adjusted Arena-Hard win rate (WR).

Supplementary material. The supplementary package includes TFPO training code, evaluation utilities, canonical-generation and bootstrap schemas, an offline stability reanalysis entry point, and the gold-span annotation protocol and worker disclosure.

C Robustness to a Larger Preference Mixture

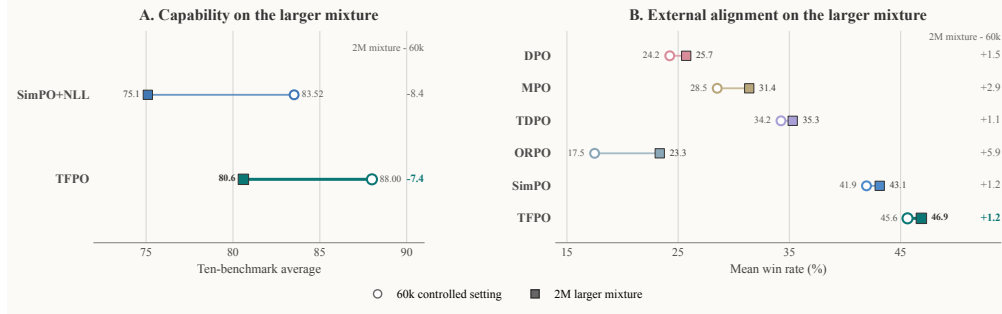


Figure 4: Robustness to a larger 2M preference mixture. Left: matched ten-benchmark averages for SimPO+NLL and TFPO in the controlled 60k setting and under the larger mixture. Right: external-alignment composites for the same settings, computed from AlpacaEval 2.0 LC/WR and Arena-Hard-v2.0-Preview WR across Qwen, Llama, and Mistral. Because data composition also changes, the connecting lines summarize robustness rather than a controlled scaling law.

Table 7: Matched Qwen3-8B capability summary under the larger preference mixture. The 60k and 2M columns use the same non-thinking evaluation; the 2M study changes both dataset size and composition.

Method	60k avg.↑	Larger-mixture avg.↑	Parser-invalid↓
SimPO+NLL	83.52	75.1	≤ 1.4%
TFPO	88.00	80.6	≤ 1.4%
TFPO gap	+4.48	+5.5	—

Figure 4 and Table 7 show that the main conclusion survives the changed data regime: TFPO remains 5.5 points above the NLL-anchored control, with parser-invalid outputs at or below 1.4%. This supports larger-mixture robustness at the aggregate level, not per-task effects, parameter scaling, or a controlled data-scaling curve.

External alignment under the larger mixture. Table 8 reports judge-based evaluation under the same larger preference mixture. The pattern is consistent with the controlled 60k setting: TFPO remains strongest across all three model families and all reported external-alignment metrics.

Table 8: External alignment under the larger 2M preference mixture. AlpacaEval 2.0 uses LC/raw WR; Arena-Hard-v2.0-Preview reports its official control-adjusted WR.

Method	Qwen3-8B			Llama-3.1-8B-Instruct			Mistral-7B-Instruct-v0.3		
	AlpacaEval 2.0		Arena-Hard	AlpacaEval 2.0		Arena-Hard	AlpacaEval 2.0		Arena-Hard
	LC	WR	WR	LC	WR	WR	LC	WR	WR
SFT/Instruct	19.8	31.2	34.1	7.1	23.1	27.5	11.3	19.2	13.8
DPO	26.5	35.8	40.3	12.4	28.4	32.4	14.6	23.2	17.9
MPO	35.1	41.7	46.5	25.1	29.8	35.1	19.1	26.5	23.6
SimPO	46.3	52.4	52.6	39.1	43.2	44.1	34.1	40.1	36.2
TDPO	36.7	42.6	46.0	31.1	37.6	36.3	27.2	34.6	25.6
ORPO	25.8	26.1	36.2	19.8	21.2	30.6	15.2	16.8	18.4
TFPO	49.1	57.2	56.7	43.7	49.1	49.9	35.2	43.0	37.9

D Ablations and Training Diagnostics

Secondary routing diagnostics. Table 9 complements the strict position-only comparison in Table 4. These rows directly evaluate where a route places its mass against the same blind gold spans. Constant, random, and shuffled assignments all preserve some aspect of the gate budget while breaking token–content correspondence. The *content+position* variant reads both hidden-state content and normalized position, so it is an auxiliary-feature diagnostic rather than a strict position-only control.

Table 9: Secondary routing diagnostics on the 1,000-response blind gold-span evaluation. The strict pure-position linear/MLP rows are the controls used in the main claim; content+position is not a position-only baseline. Bold marks the best learned route.

Route / score	Construction	Gold AUPRC↑
Constant-g	Same route mass at every response position	0.18
Matched-ratio random	Same per-response budget, randomly assigned tokens	0.18
Shuffled learned route	Same learned gate-value multiset, permuted token locations	0.24
Matched-tail heuristic	Fixed tail allocation with matched budget	0.42
Pure-position linear	Normalized position only	0.50
Pure-position MLP	Position/length features only; strongest strict-position control	0.55
Content+position	Hidden-state content plus normalized position	0.83
TFPO content gate	Hidden-state content; no explicit position feature	0.85

Table 10: Regularizer-specific diagnostics. Each ablation is evaluated on the failure mode its removed term is designed to control. The one-token-per-branch safeguard is retained in every row.

Variant	Gold AUPRC↑	Budget violation↓	Transitions/100↓	Intra-span TV↓	Early saturation↓
Default	0.85	1.6%	3.1	0.022	6.4%
w/o $\mathcal{L}_{\text{len}}$	0.56	36.1%	6.3	0.049	26.9%
w/o TV	0.68	3.5%	15.8	0.126	11.4%
w/o entropy	0.59	17.1%	8.9	0.071	52.1%
w/o all regularizers	0.42	49.2%	18.2	0.151	66.7%

What each term controls. Table 10 isolates the failure mode targeted by each regularizer. Early saturation is the fraction of tokens with $g < 0.05$ or $g > 0.95$ at 10% training progress. Removing $\mathcal{L}_{\text{len}}$, with the shared one-token-per-branch safeguard retained, increases budget violations from 1.6% to 36.1%; removing TV increases assignment transitions from 3.1 to 15.8 per 100 tokens and intra-span TV from 0.022 to 0.126; removing entropy raises early saturation from 6.4% to 52.1%. Figure 5 visualizes these three intended effects. No default seed produced non-finite values or branch-budget collapse; these diagnostics are empirical rather than a convergence guarantee.

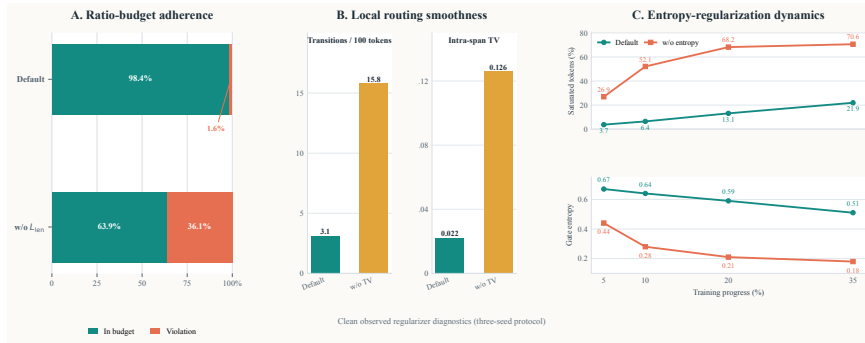


Figure 5: Regularizer-specific diagnostics. A: 98.4% of default responses satisfy the gate budget versus 63.9% without $\mathcal{L}_{\text{len}}$. B: removing TV increases transitions/100 tokens and intra-span TV from 3.1/0.022 to 15.8/0.126. C: without entropy regularization, 52.1% of tokens are saturated at 10% training progress, versus 6.4% by default.

E Full Mechanistic Evaluation Results

Final-answer stability. Table 11 expands the compact scorecard in Table 5. The same matched generation records drive every metric, with aggregation performed prompt $\rightarrow$ task $\rightarrow$ training seed. TFPO improves agreement and majority accuracy on BBH, GSM8K, and MATH while also increasing explanation diversity; the stability result is therefore not driven by one task or by repetitive templating.

Table 11: Per-task repeated-sampling stability, averaged over three training seeds. Each prompt has $K = 4$ aligned non-thinking samples; missing/invalid answers remain in the denominator. Overall is the task macro-average.

Split	Method	Agreement@4 $\uparrow$	MajorityAcc@4 $\uparrow$	AvgAcc@4 $\uparrow$	Div.@4 $\uparrow$	Valid $\uparrow$
Overall	SimPO	88.33	80.69	85.73	59.01	99.67
	SimPO+NLL	89.01	81.69	86.10	59.72	99.70
	TFPO	92.69	86.32	90.57	62.33	99.80
BBH	SimPO	84.03	80.03	80.50	59.03	99.70
	SimPO+NLL	84.40	80.47	80.50	60.00	99.70
	TFPO	88.00	85.03	86.00	61.97	99.80
GSM8K	SimPO	94.97	89.97	92.40	61.03	99.70
	SimPO+NLL	95.60	90.50	92.50	61.00	99.70
	TFPO	99.07	93.97	96.10	64.03	99.80
MATH	SimPO	86.00	72.07	84.30	56.97	99.60
	SimPO+NLL	87.03	74.10	85.30	58.17	99.70
	TFPO	91.00	79.97	89.60	61.00	99.80

Seed robustness, a MATH-500 bridge, and cross-family transfer. Table 12 separates three kinds of robustness. Panel A shows that the agreement gain is positive for every training seed; its confidence intervals come from two independent 10,000-replicate paired-prompt bootstraps. Panel B repeats the stability test at $K = 8$ on standard MATH-500. Panel C reports the three-seed external-alignment composite behind the cross-family statement in the main text.

Table 12: Robustness of the stability and cross-family results. External composite is the mean of AlpacaEval 2.0 LC/raw WR and Arena-Hard-v2.0-Preview control-adjusted WR. Values after $\pm$ are sample standard deviations over three training seeds; displayed seed gaps are rounded, while means and standard deviations use the underlying per-seed estimates.

A. Agreement@4 gaps						
Comparison	Seed gaps	Mean $\pm$ std	Paired 95% CI	MajAcc gap	Div. gap	
TFPO – SimPO	3.90/4.63/4.53	+4.36 $\pm$ 0.40	[3.06, 5.66]	+5.63	+3.32	
TFPO – SimPO+NLL	3.13/3.93/3.97	+3.68 $\pm$ 0.47	[2.38, 4.98]	+4.63	+2.61	

B. Standard MATH-500, non-thinking $K = 8$						
Method	AvgAcc@8 $\uparrow$	Agreement@8 $\uparrow$	MajorityAcc@8 $\uparrow$	Div.@8 $\uparrow$	Valid $\uparrow$	
Qwen3-8B	87.25	91.75	95.8	64.25	99.7	
SimPO+NLL	86.0	91.0	95.1	63.4	99.7	
TFPO	90.4	94.5	97.2	65.3	99.8	

C. Three-seed external-alignment composite				
Backbone	SimPO	TFPO	Gap	
Qwen3-8B	48.0 $\pm$ 0.4	52.7 $\pm$ 0.4	+4.7	
Llama-3.1-8B-Instruct	41.7 $\pm$ 0.3	45.6 $\pm$ 0.4	+3.9	
Mistral-7B-Instruct-v0.3	36.1 $\pm$ 0.3	38.6 $\pm$ 0.3	+2.5	

Functional token interventions. RR-Preserve asks whether the answer survives after low-score reasoning tokens are removed; CR-Failure asks whether deleting high-score answer tokens breaks it. Table 13 tests this signature without a shared TFPO segmenter: strict-position routers use their own scores, equal-budget comparisons remove the same token fraction, and the last panel uses external gold spans on each method’s own responses. TFPO beats the stronger position MLP on both metrics at all five thresholds and all four deletion budgets.

MATH-500 protocol and length audit. Table 14 documents the bridge between our main MATH column and the standard MATH-500 reference used in Qwen3 reporting. The main table uses

Table 13: Threshold, equal-token-budget, and method-independent gold interventions. Every learned router uses its own score; gold interventions use external spans and therefore do not assign a TFPO route to sequence-level baselines.

<i>A. Fixed-threshold sweep</i>						
Threshold	Pure-position linear		Pure-position MLP		TFPO	
	RR↑	CR↑	RR↑	CR↑	RR↑	CR↑
0.3	73.7	33.3	77.2	38.5	91.1	62.7
0.4	69.9	29.6	73.4	34.7	86.2	55.9
0.5	66.5	27.4	69.8	31.5	81.98	47.94
0.6	60.2	23.9	63.7	27.6	78.3	42.1
0.7	54.1	18.7	57.2	22.1	72.8	36.5

<i>B. Equal-removal-budget sweep</i>						
Removed tokens	Pure-position linear		Pure-position MLP		TFPO	
	RR↑	CR↑	RR↑	CR↑	RR↑	CR↑
10%	78.6	24.7	82.2	29.5	94.2	47.9
20%	72.9	35.5	76.5	40.2	89.3	62.8
30%	65.4	46.1	69.2	51.0	82.1	75.2
40%	58.5	58.3	62.4	63.2	74.0	84.1
Mean	68.9	41.2	72.6	46.0	84.9	67.5

<i>C. Method-independent gold-span intervention</i>				
Method / router	Reasoning preserve↑	Answer failure↑	Joint geom.↑	
DPO+NLL	69.2	54.7	61.5	
SimPO+NLL	71.9	57.1	64.1	
TFPO (pure-position linear route)	73.1	59.1	65.7	
TFPO (pure-position MLP route)	76.5	63.8	69.9	
TFPO (content-gated route)	88.6	80.4	84.4	

EleutherAI/hendrycks_math; the bridge uses all 500 examples of the standard MATH-500 test split and mode-specific Qwen3 sampling. With prompts and sampling indices fixed, varying only the output cap confirms that 512 tokens is restrictive for every method. At 32,768 tokens, truncation is negligible and TFPO remains above SimPO+NLL in both modes. Across the full ten-task capability suite, overall truncation, missing-answer, and parser-invalid rates are at most 0.1%, 0.4%, and 0.5%, respectively, with a maximum per-task failure rate of 0.8%.

Table 15 records the full population, annotation, recovery, bootstrap, and position audit behind the main gold-span comparison.

Gold-span annotation protocol and worker information. The annotation rules and worker disclosure are provided in the supplementary file reproducibility/gold_span_annotation_protocol.md. The labels were produced by two employees of an external annotation company in mainland China. The rate was RMB 10 per annotation item, and both annotators agreed that their annotations could be used for research. Both annotators were blind to model identity and all gate scores. The 300 double-annotated responses were labeled independently by both annotators before disagreements were adjudicated; before adjudication, Cohen’s κ was 0.92, span IoU was 0.89, and the median boundary disagreement was one token.

The candidate order and structural exclusion rules were fixed before inspecting gate values, correctness, extraction outcomes, or downstream metrics. Integrity checks used only candidate IDs,

Table 14: MATH-500 model/mode and output-length audit. AvgAcc@8 averages correctness over eight aligned samples; empirical pass@8 is the fraction of problems with at least one correct sample. Missing denotes an untruncated output for which the frozen extractor finds no answer. Fine-tuned values are equal-weight three-training-seed means.

A. Mode-matched MATH-500 bridge at max new tokens 32,768						
Mode	Method	AvgAcc@8↑	Pass@8↑	Trunc.↓	Missing↓	Median/p95 tokens
Non-thinking	Qwen3-8B	87.25	99.8	0.1%	0.3%	281/953
	SimPO+NLL	86.0	99.6	0.1%	0.3%	272/850
	TFPO	90.4	99.9	0.1%	0.2%	255/725
Thinking	Qwen3-8B	97.2	100.0	0.2%	0.2%	502/4,203
	SimPO+NLL	97.0	99.9	0.2%	0.2%	481/4,502
	TFPO	97.5	100.0	0.1%	0.1%	433/3,703

B. Length sensitivity with aligned prompts and sampling indices				
Mode	Method	AvgAcc@8: 512/4096/32768↑	Trunc.: 512/4096/32768↓	Missing: 512/4096/32768↓
Non-thinking	Qwen3-8B	82.0/87.2/87.25	18.2/0.4/0.1%	3.0/0.4/0.3%
	SimPO+NLL	82.5/85.9/86.0	14.3/0.3/0.1%	2.5/0.4/0.3%
	TFPO	86.0/89.9/90.4	10.1/0.8/0.1%	1.5/0.2/0.2%
Thinking	Qwen3-8B	76.0/92.0/97.2	48.5/5.4/0.2%	8.0/1.1/0.2%
	SimPO+NLL	77.0/90.8/97.0	45.7/6.6/0.2%	7.0/1.2/0.2%
	TFPO	81.0/93.8/97.5	38.3/4.2/0.1%	4.0/0.5/0.1%

Table 15: Gold-span population, annotation, recovery, and position audit. AUPRC pools token labels/scores over responses; each paired bootstrap replicate resamples responses and recomputes the pooled metric, retaining valid all-other responses.

A. Prespecified, model- and score-blind population audit	
Record status	Count
Candidate responses	1,018
Exact duplicate removed	9
Irrecoverable token/gate alignment mismatch removed	5
Corrupted/incomplete serialization removed	4
Removed using correctness, extraction, identity, score, or downstream metric	0
Final population (five tasks × 200)	1,000

B. Annotation audit before adjudication	
Item	Value
Double-annotated responses	300
Cohen’s κ	0.92
Span IoU	0.89
Median boundary disagreement	1 token
Answer-bearing token prevalence	0.18

C. Natural-response recovery and paired response-bootstrap differences				
Router	AUPRC↑	Precision↑	Recall↑	F1↑
Pure-position MLP	0.55	0.63	0.57	0.60
TFPO content gate	0.85	0.84	0.80	0.82
TFPO – MLP (95% CI)	+0.30 [0.27, 0.33]	+0.21 [0.18, 0.24]	+0.23 [0.20, 0.26]	+0.22 [0.19, 0.25]

D. Content-preserving answer-position counterfactual				
Router / score	Answer first	Answer middle	Answer last	Max gap↓
Matched-tail heuristic	0.18	0.38	0.68	0.50
Pure-position linear	0.25	0.42	0.62	0.37
TFPO content gate	0.83	0.84	0.85	0.02

serialization status, and token/offset/gate-array consistency, and were applied once to the shared candidate manifest before any router comparison. Incorrect responses with an attempted answer span were retained; valid responses without one were retained with all response tokens labeled `other`. In the counterfactual, reasoning and answer content are held fixed while the answer is moved to the beginning, middle, or end.

F MLLM Evaluation Details and Qualitative Examples

Table 16 gives the per-benchmark breakdown behind the aggregate MLLM result in Section 4.6. The multimodal runs use the processed MM-RLHF and OpenBMB RLAI-F-V preference mixture for one epoch with global batch size 16 and maximum sequence length 4096; the ViT and aligner remain frozen. The reported multimodal TFPO run uses $\beta = 0.05$ and $\lambda_{\text{null}} = 0.2$. We evaluate the MathVista_MINI, MathVision, MMMU_DEV_VAL, AI2D_TEST, MMVet, LLaVABench, MMBench, MMStar, POPE, and HallusionBench splits. We report only the judge-based gpt-4o-mini MLLM scores in the main text, because strict exact matching underestimates many free-form multimodal answers.

Table 16: Per-benchmark multimodal results for preference methods trained from Qwen3-VL-8B-Instruct. Scores are judge-based with gpt-4o-mini.

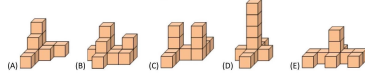
Method	Avg.	Reasoning			General VQA					Hallucination	
		MathVista	MathVision	MMMU	AI2D	MMVet	LLaVA-Bench	MMBench	MMStar	POPE	HallusionBench
Qwen3-VL-8B-Instruct	66.74	69.20	15.00	52.67	80.57	49.68	88.80	84.70	64.87	87.98	73.92
DPO	66.86	71.30	17.07	48.67	81.99	50.64	88.30	84.02	65.87	87.90	72.87
MPO	67.54	67.60	20.49	52.11	85.07	53.62	85.70	85.57	65.73	88.00	71.50
SimPO	52.16	55.40	12.96	38.44	61.14	37.25	69.20	70.57	45.33	84.20	47.11
TDPO	63.86	59.88	26.34	48.12	80.12	50.89	81.20	82.50	59.28	81.26	69.00
ORPO	19.52	23.60	5.07	0.56	3.34	73.39	10.00	0.11	0.00	26.60	52.58
TFPO	71.03	70.90	30.93	55.67	85.90	59.22	86.30	87.28	69.20	90.35	74.50

Figure 6 presents three nontrivial cases in which TFPO is correct and parseable while a representative baseline shows format drift, a wrong option, or excessive verbosity. They span complementary final-answer instabilities and are qualitative illustrations, not aggregate evidence.

Visual input

Case 1: Visual counting.

MathVision, dice solids



Failure mode: format drift.

Question.

Max has 10 dice. Which one of the five solids can he build? The prompt asks for a final option letter.

TFPO raw excerpt.

“To determine which solid Max can build with 10 dice, we need to count the number of dice in each option: (A) The solid has 10 dice. (B) The solid has 11 dice. (C) The solid has 10 dice. (D) The solid has 11 dice. (E) The solid has 10 dice. Max has 10 dice, so he can build solids (A), (C), or (E). However, the question asks for the correct option letter, and the answer is: A.”

ORPO raw excerpt.

“After counting the cubes, the fifth solid uses exactly 10 dice. Therefore, Max can build the fifth figure.”

Interpretation.

This example separates visual reasoning from answer-field discipline. ORPO gives a semantically close description but omits the required option letter; TFPO keeps a parseable final answer field.

Case 2: Map reasoning.

MMMBench, Silk Road map



Failure mode: wrong answer.

Question.

The map shows Silk Road routes around 1300 CE. Which statement is true: connected East Asia/Middle East/Europe, connected East Asia and the Americas by sea, or only land routes?

TFPO raw excerpt.

“A. The Silk Road connected parts of East Asia, the Middle East, and Europe. This answer is correct based on the map shown. The map clearly illustrates the extensive network of overland routes (shown in dark lines) that connected East Asia with the Middle East and Europe. It also shows some sea routes (in pink) that connected different parts of the network.”

TDPO raw excerpt.

“The map shows trade routes extending across oceans and connecting distant continents. Since the routes appear to cross the Pacific Ocean, the Silk Road connected East Asia and the Americas by sea. Answer: B.”

Interpretation.

Here the failure is not formatting but content. TDPO produces a plausible-sounding explanation that conflicts with the map, while TFPO keeps the visual conclusion and option letter aligned.

Case 3: Numerical VQA.

MMMU, surveying problem

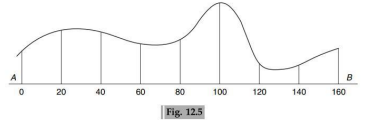


Fig. 12.5

Point	A									B
Distance (m)	0	20	40	60	80	100	120	140	160	
Offset length (m)	23	40	42	30	32	60	10	14	22	

Failure mode: excessive verbosity in a strong baseline.

Question.

Use the offsets in the figures to compute the area between line AB and a meandering stream by both the trapezoidal rule and Simpson’s rule.

TFPO raw excerpt.

“The offsets are: 23, 40, 42, 30, 32, 60, 10, 14, 22. The interval is $h = 20$ m. Using the trapezoidal rule: $\text{Area} = h/2 \times [\text{first offset} + \text{last offset} + 2 \times \text{sum of intermediate offsets}] = 5010 \text{ m}^2 = 0.5010 \text{ hectares}$. Using Simpson’s rule: $\text{Area} = h/3 \times [y_0 + y_8 + 4(y_1 + y_3 + y_5 + y_7) + 2(y_2 + y_4 + y_6)] = 5260 \text{ m}^2 = 0.5260 \text{ hectares}$. Therefore, the correct answer is: C.”

SimPO raw excerpt.

“This problem asks us to calculate the area between a straight baseline AB and a meandering stream. Since the offsets are taken at equal intervals, numerical integration methods can be used. The distance interval is 20 m, and the offsets are 23, 40, 42, 30, 32, 60, 10, 14, and 22 m. For the trapezoidal rule, the first and last offsets are taken once, while the intermediate offsets are doubled. This gives an area of 5010 square meters, which is equal to 0.5010 hectares. For Simpson’s rule, because there are eight equal intervals, the rule can be applied. The odd-positioned offsets are multiplied by 4, and the even-positioned intermediate offsets are multiplied by 2. Carrying out the calculation gives 5260 square meters, which is 0.5260 hectares. Since option C contains these two values, the answer is C.”

Interpretation.

SimPO is a strong baseline and reaches the right answer, but its raw response is much longer than necessary for an option-selection task. This case illustrates why we care about stable, concise final-answer anchoring rather than only answer correctness.

Figure 6: Qualitative multimodal examples. We show the visual input and response excerpts for TFPO and representative baselines. The three cases highlight complementary final-answer instability modes behind the multimodal results in Section 4.6: missing the required answer format, choosing an unsupported option, and producing an unnecessarily long answer despite reaching the correct conclusion.

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Justification: The gold-span evaluation used two paid annotators employed by an external annotation company in mainland China. The rate was RMB 10 per annotation item, and both annotators agreed that their annotations could be used for research. Appendix E and the supplementary file `reproducibility/gold_span_annotation_protocol.md` provide the operational labeling instructions, blinding, and the independent double-annotation and subsequent adjudication procedure for the 300 double-annotated responses.

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